Balancing strategies

The payoff that an individual animal gains—or the price it risks paying—by displaying a particular behavior can be quantified, so biologists can work out which strategies are likely to be most stable by using mathematical models (see The hawk-dove “game”). If the model does not match the behavior of animals in the real world, then it suggests that stability has not evolved.

In real rather than hypothetical ecosystems, it is not a single strategy that is stable, but the balance between two or more strategies within the system as a whole. The overall balance is therefore better called an evolutionarily stable state, and not a strategy. Such a balance emerges when all individuals have equal fitness: they pass on their genes to the same extent. The state remains stable, even when there are minor changes in the animal’s environment.